Sliding Window Attention with Recurrent Memory

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Abstract. We show that sliding window attention combined with a compact recurrent memory keeps long-context quality while reducing cost. On three long-document benchmarks it matches full attention within one point at 2.1× the speed.

1 Introduction

Long documents challenge transformer models because full attention grows quadratically with sequence length. Practitioners therefore truncate inputs, which discards evidence that later questions depend on.

The effect holds across model sizes from 125M to 1.3B parameters. We keep the tokenizer and optimiser settings of the baseline unchanged. Throughput is measured on a single accelerator with a batch of eight documents. Each result is the mean of three seeds; standard deviations are below 0.4 points.

We study sliding window attention with a small recurrent memory. Each layer attends to a fixed window of recent tokens and to a compressed memory carried across windows.

We keep the tokenizer and optimiser settings of the baseline unchanged. Throughput is measured on a single accelerator with a batch of eight documents. Each result is the mean of three seeds; standard deviations are below 0.4 points. Shorter windows lower memory use but hurt questions whose evidence is far away.

Our contributions are threefold: a training recipe that keeps the window small, an analysis of what the memory stores, and an evaluation on three long-document benchmarks.

Throughput is measured on a single accelerator with a batch of eight documents. Each result is the mean of three seeds; standard deviations are below 0.4 points. Shorter windows lower memory use but hurt questions whose evidence is far away. The memory has 64 slots of the model dimension and is reset at document boundaries.

2 Related Work

Sparse attention patterns reduce cost by restricting which tokens interact. Recurrent memories, by contrast, summarise the past into a fixed number of slots.

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model dimension and is reset at document boundaries. Gradients flow through the memory for two windows and are truncated beyond that.

Retrieval-augmented models fetch passages from an external index; they are complementary to our approach.

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3 Method

Given a sequence split into windows of w tokens, layer l computes attention over the current window and the memory state m from the previous window. The memory is updated with a gated write.

Gradients flow through the memory for two windows and are truncated beyond that. Inspection of the memory shows that it keeps entity names and numbers most often. We release the training configuration so that the results can be reproduced. Documents longer than 32k tokens are rare in the benchmarks and are left out.

3.1 Memory update

The gate decides how much of each memory slot to overwrite. We initialise the gate bias so that the memory is mostly preserved early in training.

Documents longer than 32k tokens are rare in the benchmarks and are left out. The auxiliary loss weight is 0.1 in all runs; larger weights slow convergence. At inference time the memory adds less than three percent to the latency. The effect holds across model sizes from 125M to 1.3B parameters.

3.2 Training objective

We train with the standard language-modelling loss and an auxiliary reconstruction loss on the memory.

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4 Experiments

We evaluate on three benchmarks of long documents: question answering, summarisation and retrieval. Sliding window attention with memory matches full attention within one point while running 2.1 times faster.

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Table 1 reports accuracy and throughput. Ablations show the memory matters most for questions whose evidence lies more than four windows back.

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Model	QA	Summ.	Tokens/s
Full attention	71.2	38.4	1.0×
Sliding window	64.0	35.1	2.4×
Ours	70.5	38.0	2.1×

5 Limitations

All corpora are English. We did not ablate the window size beyond three values, and the memory size is fixed.

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6 Conclusion

A small recurrent memory lets sliding window attention keep long-range evidence at a fraction of the cost of full attention.

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